Student ID:_____, Family Name:_____, Given Name:_____

COMP 212 Computer Architecture

Mid-term Exam

Fall 2008

To be fair, please do NOT open the exam book, until told so.

Notice:

Mid-term is **close book, close notes, NO calculator and NO discussions**. Please write down the details of your solutions, partial results will be given partial credits. Don't rush, you should have plenty of time, do a careful job especially with binary computing and conversion.

Add blank papers for solutions if you need. Write down your name and student id on every page, in case the exam papers get mixed up.

Good Luck !

Useful Info:

Conversion Code - Chart															
DECIMAL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
BINARY	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111

Grades:

Total is 75 pts

Q1:_____	+ Q2:_____	+ Q3:_____	+ Q4:_____	+ Q5:_____	+ Q6:_____	= Total: _____
10	10	10	15	10	20	75

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Q1: _____/10pts

Q1. [10 pts, 1 pt each] General Knowledge of Computer, True or False (fill in T or F)

1. (__) Register is implemented inside the main memory on computer board
2. (__) von Neumann architecture has data and instructions in the same memory space.
3. (__) Static RAM is typically used to implement Cache
4. (__) Dynamic RAM is faster than Static RAM in access time.
5. (__) In modern computers, CPU and Memory are connected by BUS
6. (__) Direct Mapping in Cache system is more complex in implementation than Set Associative Mapping
7. (__) Floating point data points are uniformly represented on the real line.
8. (__) unsigned integer data points are uniformly represented on the real line.
9. (__) Read and write operations can be performed on EPROM devices
10. (__) The density of data on the hard disk does not affect data access speed.

Answers: F T T F T F F T T F

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Q2: _____/10pts

Q2. [10 pts] Number and Arithmetic – Un-Signed Integers binary representation and arithmetic

1. [4 pts] Give the 16-bit binary representation of the following numbers:

A= 118 = (0000 0000 0111 0110)$_2$

B= 98 = (0000 0000 0110 0010)$_2$

C= 46 = (0000 0000 0010 1110)$_2$

D= 15 = (0000 0000 0000 1111)$_2$

2. [6 pts] Compute the following problems in binary form :

A – B = 0000 0000 0001 0100

C x D= 0000 0010 1011 0010

A / D = 111, reminder: 1101

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Q3: _____/10pts

Q3. [10 pts] Number and Arithmetic - Signed Integers, two's complement representation

1.[2 pts] Let the decimal numbers A=54, B= -77, give their 8-bit 2's complement representation :

A =54= 0011 0110

B=-77= $[01001101]_b$ + 1=10110010 +1 = 10110011

2. [2 pts] Compute A + B in 2's complement:

```
    0011 0110
    1011 0011
-----------------------
    1110 1001
```

A+B = 1110 1001 = -23.

3. [3 pts] Compute A-B in 2's complement, is it overflow ? why or why not ?

```
 -B = 77 = 0100 1101
    0011 0110
+   0100 1101
--------------------------
    1000 0011, overflow, MSB different
```

4. [3 pts] Give the 16-bit 2's complement representation of A, and B. Compute A-B. Is it overflow ?

A= 0000 0000 0011 0110

-B=0000 0000 0100 1101

A-B = 0000 0001 0000 0011, no overflow

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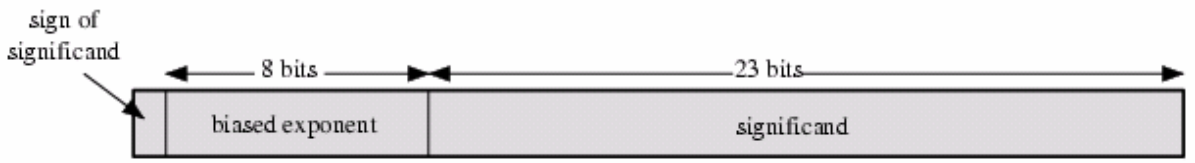
Q4:

_____/15pts

Q4. [15 pts] Floating point numbers. For the 32 bit Floating point system given below, the exponent bias is 127, i.e., if the true exponent is 5, the biased exponent would be 132.



Find the sign bit, exponent, and significand representations of the following numbers, please give steps and intermediate results also. You only need to provide up to 6 digits for the fractional precision part.

1. [5 pts] 118.75

118 = 0111 0110, 0.75 = 0.11
Sign = 0
Exponent = 111 0110.11 = 1.11011011 x 2^6, 6+127=133=1000 0101
Significand = 11011011 0000000000000

2. [5 pts] $-(\frac{1}{32}+\frac{1}{128})$

5/128=5 x 2^{-7}= 1.01 x 2^{-5}
Sign=1, exponent = 127-5=122=0111 1010, significand = 010000000000000

3.[5 pts] 0.15

0.15 = 0.00100110011001100110011……
= 1.00110011001100110011…. x 2^{-3}
Sign=0, exponent=-3+127=124=0111 1100, significand = 001 1001 1001 1001 1001 …..

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Q5:

_____/10pts

Q5. [10 pts] Consider a disk system, which has a track seek time of t_{seek} =10ms (mini-second). The disk rotation speed is ***r*** = 9000 rpm (round per minute), and each track on the disk has ***S***=600 sectors, each sector has total ***B***=512 bytes data.

1. [5 pts] What is the average time it takes to read 1024 bytes data ?

$T(X) = (t_{seek} / 1000) + 1/(2*(r / 60)) + X/(S*B*(r / 60))$ (sec)

T(1024) = 13.355 ms

2. [5 pts] If the sector density is increased to ***B***=1024 bytes, and rotation speed is increased to 12000 rpm, what is the average time it takes to read 1024 bytes data ?

T = 12.508 ms

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Q6: _____/20pts

Q6. [20pts] Cache and Memory System

A memory system has 16M bytes. The memory is organized into blocks of 64bit/8 bytes each, and the cache has total 512K bytes, organized into cache lines of 8 bytes each:

1. [2pts] How many bits are needed to address all bytes ?

.

24 bits

2. [2pts] How many memory blocks we have ? How many bits are needed to address all memory blocks ?

.

2M blocks, 21bits

3. [2 pts] How many cache lines we have ? how many bits are needed to address all cache lines ?

.

64K cache lines, 16 bits.

4. [6 pts] For direct cache mapping, what would be the Block address, Tag, Line and Word in Hex form for the following byte addresses ?

Address	Block Addr	Tag	Line	Word
A1 = DB86E2h	1B70DC	T_1=1B	L_1=70DC	W_1=2
A2 = 8B86E1h	1170DC	T_2=11	L_2=70DC	W_2=1

A1 = DB86E2=110110111000011011100010,

A2 = 8B86E1=100010111000011011100001,

8 byte per block, so *Word* should be 3 bits, 16bits for *Line* and 24-(3+16)=5bits for *Tag*. Therefore:

A1: 11011, 0111000011011100, 010 =>

Block= 1B70DCh, Tag=1Bh, Line=70DCh, Word=2h

A2: 10001, 0111000011011100, 001=>

Block= 1170DCh, Tag=11h, Line=70DCh, Word=1h.

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5. [6 pts] If cache is organized into 4-way set associative cache, how many bits are required for the Tag, Set and Word ? What are the Tag, Set and Word values for the following byte addresses ?

Address	Tag	Set	Word
A1 = DB86E2h	T_1=6D	S_1=30DC	W_1=2
A2 = 8B86E1h	T_2=45	S_2=30DC	W_2=1

We have 64K cache lines, 4-way associative, so we have 64K / 4 = 16K cache set, each with 4 cache lines.
16K => we need $2^4 x 2^{10} = 2^{14}$ cache sets. So we need 14 bits for cache set address.
3 bits for word selector, 14 bits for cache set, that leaves 7 bits for Tag.
A1: 1101101, 11000011011100, 010 =>
Block= 1B70DCh, Tag=6Dh, Set=30DCh, Word=2h
A2: 1000101, 11000011011100, 001=>
Block= 1170DCh, Tag=45h, Set=30DCh, Word=1h.

6. [2 pts] For the 4-way set associative cache system discussed in Q6.4, if Cache Set at set address 30DCh has the following 4 cache line content and Tags:

Tag	Content
1000101	00FF000000FF00FF

Tag	Content
1000001	FFFFFFFFFFFFFF

Tag	Content
10110101	00000000000000000

Tag	Content
1000111	0AAAAAAAAAAA

If CPU request memory byte at addresses A1 = DB86E2h, will this be a cache hit or miss ? How about for memory request at 8B86E1h, will this be a cache hit or miss ?

A1 a cache miss, because A1's tag: 1101101 not in cache
A2 a cache hit, because A2's tag: 1000101, in cache.